

CoRoT-2b

R-0828

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_260623 · created 2026-07-28T16:47:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461214.7603 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.222 +/- 0.057
Transit depth (Rp/Rs)^2	4.93 +/- 2.51 [%]
Orbital Inclination (inc)	85.25 +/- 1.91
Airmass coefficient 1 (a1)	1.22 +/- 0.18
Airmass coefficient 2 (a2)	-0.129 +/- 0.031
Scatter in the residuals of the lightcurve fit is	14.27 %
Best Comparison Star	None
Optimal Aperture	6.56
Optimal Annulus	13.57
Transit Duration (day)	0.088 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.222 ± 0.057 vs 0.1667 (deviation 0.74σ)
Duration fit vs published	0.088 d vs 0.095 d (deviation 0.41σ)
Mid-time O–C	9.8 min
Depth SNR	3.0
Residual scatter	14.27 %

Light curve

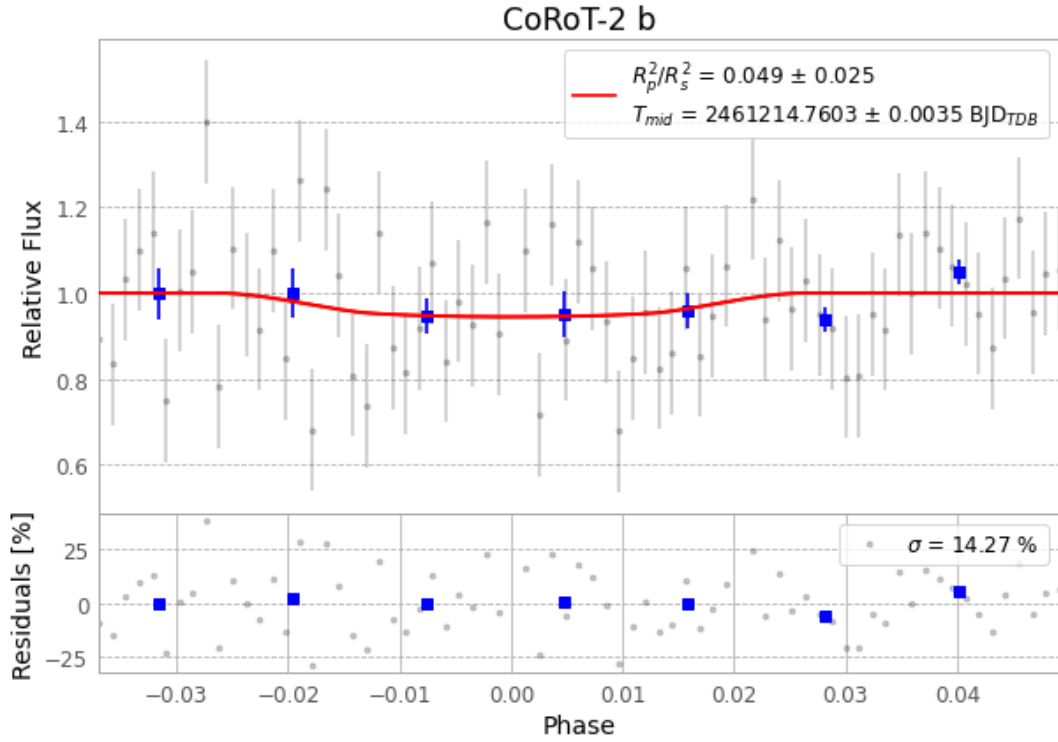


Figure 1 — FinalLightCurve_CoRoT-2 b_2026-06-22.png (SHA-256 59f8a00508572fe0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [527, 462], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75840fe3f752b204...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), CoRoT-2260623043315.FITS → CoRoT-2260623080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-260623.log (62ecdc42e164...), FinalLightCurve_CoRoT-2 b_2026-06-22.png (59f8a0050857...), FinalLightCurve_CoRoT-2 b_2026-06-22.csv (e37481a809c0...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 16:47Z.